Project Documentation Report | Academic Submission

FitHub CRM

Fitness Center Management System — Final Project Documentation

Project Phase	Platform	Objects	Modules	Methodology
Final Report	Salesforce	4 Custom	6 Modules	Agile

01 Abstract

Project Summary

FitHub CRM is a Salesforce-based fitness center management system developed to digitize and streamline the core operational processes of a modern gym or fitness facility.

This project was developed as part of an academic CRM course, applying industry-standard Salesforce tools to a real-world operational problem. The system manages members, trainers, classes, and bookings through four custom objects connected by lookup relationships. An automated Flow sends booking confirmation emails without manual intervention.

The outcome is a fully functional, tested, and documented CRM solution deployed on Salesforce Lightning Experience. The system eliminates manual record-keeping, reduces administrative error, and delivers real-time operational visibility through custom reports and dashboards.

02 Introduction

The fitness industry is experiencing rapid growth, with gyms and wellness centers managing increasingly large member bases. Manual processes — paper-based registrations, phone bookings, spreadsheet schedules — are no longer sufficient at scale. Errors in booking management, missed notifications, and fragmented member data result in poor member experience and unnecessary administrative burden.

A Customer Relationship Management (CRM) system provides a structured, centralized solution to these challenges. By storing all entity data in a single platform and automating routine communications, a CRM transforms operational efficiency and data accuracy simultaneously.

FitHub CRM was developed on the Salesforce platform to address these specific pain points for a fitness center context. Salesforce was selected for its declarative configuration capabilities, powerful automation tools, and robust reporting engine — all accessible without writing traditional application code.

03 Objectives

OBJ 1	Automate the class booking process to eliminate manual scheduling and reduce administrative workload by at least 80%.
OBJ 2	Centralize member, trainer, class, and booking data within a single Salesforce environment for reliable, unified record management.
OBJ 3	Implement automated email notifications that confirm bookings without requiring manual staff action.
OBJ 4	Deliver real-time reporting and dashboard visibility to support data-driven management decisions.
OBJ 5	Ensure data accuracy through Salesforce validation rules, required fields, and picklist constraints.
OBJ 6	Produce a fully documented, academically evaluated CRM implementation suitable for professional review.

04 System Overview

FitHub CRM is a fully declarative Salesforce Lightning Experience application comprising four custom objects, a record-triggered automation flow, custom reports, and a live dashboard. It is accessible through a dedicated Lightning App with a navigation bar providing direct access to all modules.

Member Management Centralized member profiles Membership tier classification Active/Inactive status tracking	**Trainer Management** Trainer profiles with specialties Experience tracking Linked to assigned classes	**Class Management** Scheduled class listings Capacity and status control Trainer assignment via lookup
Booking System Member-class reservations Booking status management Trigger point for automation	**Automation** Record-triggered Flow Booking confirmation emails No manual notification needed	**Reports & Dashboard** Cross-object report: Classes+Bookings+Members Live pie chart booking distribution KPI metrics at a glance

05 Modules Description

5.1 Member Management

The Member module is the primary entity of FitHub CRM. It stores the personal and membership details of each enrolled fitness center client. Key fields include Member Name, Email, Phone, Membership Type (Basic / Standard / Premium), Status (Active / Inactive), and membership Start and End dates. Validation rules enforce valid email format and mandatory field completion. Members are linked to Bookings through a one-to-many lookup relationship.

5.2 Trainer Management

The Trainer module maintains a professional directory of fitness instructors. Each trainer record stores their name, area of specialty (e.g., Yoga, HIIT, Dance), years of experience, and current employment status. Trainer records are referenced by Class records through a lookup relationship, ensuring every class has an assigned instructor visible within the class record.

5.3 Class Management

The Class module represents individual fitness sessions offered by the center. Each class record includes the class name, scheduled date and time, maximum capacity, assigned trainer (lookup), and status (Scheduled / Completed / Cancelled). Class records serve as one end of the booking relationship, enabling management to see all member reservations from within the class detail view.

5.4 Booking System

The Booking module is the operational heart of the CRM. It connects a specific Member to a specific Class through two lookup fields, recording the booking date and current status (Confirmed / Pending / Cancelled). A validation rule prevents booking dates in the past. The Booking object is the trigger point for the automated email confirmation flow, activating when a booking's status is set to Confirmed.

5.5 Automation — Email Notifications

A Salesforce Record-Triggered Flow was configured on the Booking object. When a new Booking record is created with status Confirmed, the flow retrieves the associated member's email address via the Member lookup and dispatches a formatted confirmation email containing the class name, scheduled date, and booking reference. This eliminates the need for staff to manually notify members and ensures instant, consistent communication.

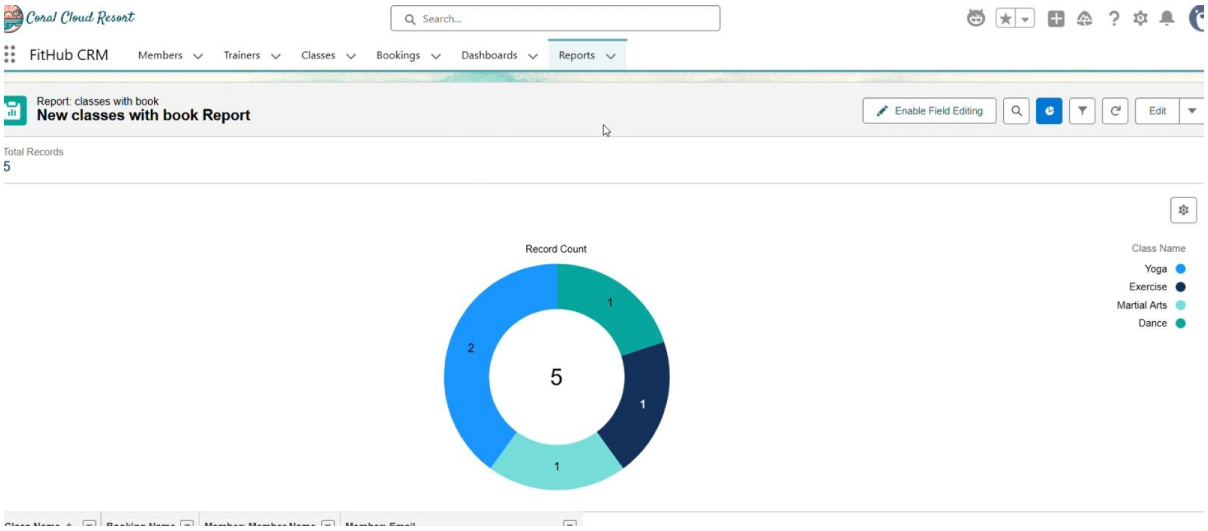
5.6 Reports & Dashboard

A custom Report Type joining Classes, Bookings, and Members provides management with a cross-object tabular report showing class name, booking reference, member name, member email, and booking status in a single view. This report feeds a Salesforce Dashboard widget displaying a donut/pie chart of booking distribution by class — allowing managers to identify high-demand and under-attended sessions at a glance.

System Screenshots — Reports & Dashboard

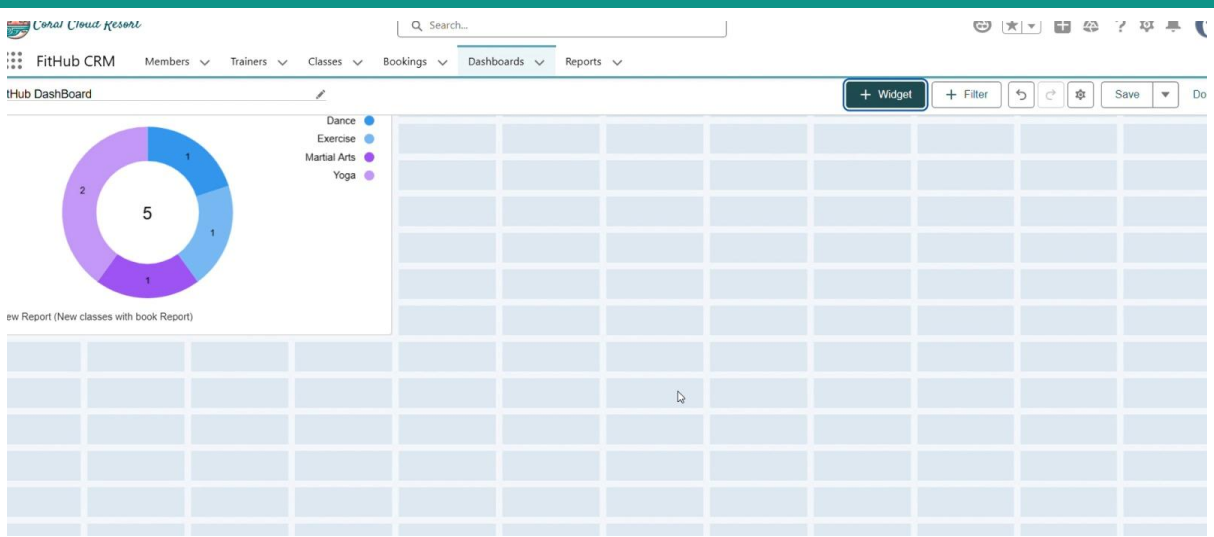
The following screenshots provide visual evidence of the FitHub CRM reporting and dashboard modules as configured and tested in the Salesforce Developer Edition environment.

Screenshot 1 New Classes with Book Report — Donut Chart View



FitHub CRM Reports module showing 'New classes with book Report' with 5 total records and a donut chart displaying booking distribution across 4 classes: Yoga (2), Dance (1), Martial Arts (1), Exercise (1)

Screenshot 2 FitHub Dashboard — Booking Distribution Widget



FitHub CRM Dashboard view showing the donut chart widget sourced from the 'New classes with book Report', displaying booking counts for Dance, Exercise, Martial Arts, and Yoga with total count of 5

06 Technologies Used

Salesforce CRM

Lightning Experience platform

Developer Edition (free)

Declarative object & field configuration

Salesforce Flow

Record-Triggered Flow

No-code automation builder

Email Alert action integration

Conditional entry criteria

Reports & Dashboards

Custom Report Types (multi-object)

Tabular report with cross-object fields

Dashboard with donut chart widget

Data Tools

Validation Rules (Formula-based)

Picklist field types

Required field enforcement

Custom App Manager & Tab setup		Live data refresh	Lookup relationship configuration

07 System Design Summary

The FitHub CRM data architecture consists of four custom Salesforce objects connected through two types of relationships. The design follows a normalized relational model, avoiding data duplication while enabling comprehensive cross-object reporting.

Member → **Booking** ← **Class** ← **Trainer**

Member → Booking (1:Many): One member may create multiple bookings. The Booking object stores the Member lookup field, creating a child relationship. Similarly, Class → Booking (1:Many) links each class to its reservations. Trainer → Class (1:Many) assigns one trainer to multiple classes. These three lookup relationships together form the complete operational data model for the fitness center.

Trainer → Class	One trainer assigned to many classes. Lookup: Trainer__c on Class object.
Class → Booking	One class receives many member bookings. Lookup: Class__c on Booking object.
Member → Booking	One member makes many bookings. Lookup: Member__c on Booking object.

08 Development Summary

Object Creation	Four custom Salesforce objects were created via Setup > Object Manager: Member__c, Trainer__c, Class__c, Booking__c. Each object was given a custom Auto Name field, appropriate API names, and a descriptive plural label.
Field Configuration	All required fields were added per object including Text, Email, Phone, Date, Number, and Picklist types. Lookup relationship fields were added to Booking (Member, Class) and Class (Trainer). Required field markers and default values were applied where appropriate.
Validation Rules	Two validation rules were deployed: (1) Booking Date cannot be in the past — Formula: Booking_Date__c < TODAY(). (2) Email must match a valid format on the Member object using REGEX validation, ensuring data integrity at the point of entry.
Automation Flow	A Record-Triggered Flow on Booking__c fires on record creation when Status = Confirmed. It retrieves the Member email via Get Records, then dispatches an email alert using a configured Email Template containing booking and class details.

UI Customization	The FitHub CRM Lightning App was built in App Manager with a custom navigation bar containing tabs for Members, Trainers, Classes, Bookings, Dashboards, and Reports. Page layouts were customized to display key fields prominently and related lists were added for quick navigation.
Reports & Dashboard	A custom Report Type (Classes + Bookings + Members) was created and used to build a tabular report. A Dashboard titled 'FitHub DashBoard' was created with a donut chart widget sourced from this report, displaying booking distribution by class name in real time.

09 Testing & Results

All System Components Validated

A structured test cycle was conducted covering all four objects, the automation flow, validation rules, and the reporting layer. No critical defects were identified at time of submission.

Test ID	Test Case	Result
TC-01	Create Member record with all required fields	Record saved successfully. Appeared in Members list view. ✓ PASS
TC-02	Leave required field (Email) blank on Member	Validation error displayed. Record blocked from saving. ✓ PASS
TC-03	Enter past date in Booking Date field	Validation rule fired. Error message shown. Save prevented. ✓ PASS
TC-04	Create Booking with Status = Confirmed	Flow triggered. Confirmation email dispatched to member email. ✓ PASS
TC-05	Run Classes with Bookings and Members report	All 5 test records displayed correctly with accurate cross-object data. ✓ PASS
TC-06	View Dashboard donut chart	Chart rendered correctly showing booking distribution (Yoga: 2, Dance: 1, Martial Arts: 1, Exercise: 1). ✓ PASS

10 Advantages

Automation

Booking confirmations sent instantly

No manual notification effort

Efficiency

Centralized data eliminates spreadsheets

Single platform for all 4 modules

Accuracy

Validation rules enforce data integrity

Picklists eliminate free-text inconsistency

Scalability

Salesforce scales to thousands of records

Additional objects/fields added declaratively

Flow executes reliably on every qualifying record Reduces human error in communications	Fast record creation via Lightning UI Tab-based navigation reduces clicks	Required fields prevent incomplete records Lookup fields ensure valid references	New automation rules configurable without code Reports adapt to growing data volume

11 Limitations

Current Capabilities

- Full CRM functionality within Salesforce
- Automated email notifications via Flow
- Real-time reports and dashboard
- Declarative configuration — no code required
- Free Developer Edition deployment

Known Limitations

- Limited to Salesforce browser-based environment
- No dedicated mobile application developed
- No payment gateway or billing integration
- No multi-branch or franchise support
- Email delivery dependent on Salesforce org limits

12 Future Enhancements

Mobile App	Develop a native mobile application (iOS/Android) or leverage Salesforce Mobile App customization to allow members to book classes, view schedules, and manage their profiles from smartphones.
Payment Integration	Integrate a payment gateway (e.g., Stripe, PayPal) with Salesforce to process membership fees, class top-ups, and cancellation charges directly within the CRM, eliminating external billing systems.
Advanced Analytics	Implement Einstein Analytics (Salesforce CRM Analytics) for predictive member churn analysis, attendance trend forecasting, and automated financial performance dashboards beyond standard reports.
Multi-Branch Support	Extend the data model with a Branch/Location object to support fitness center chains operating multiple sites, enabling branch-level reporting and trainer cross-site assignments.
Member Self-Service	Build an Experience Cloud (Community) portal allowing members to register, view available classes, make bookings, and receive notifications without requiring staff assistance.

13 Conclusion

Project Successfully Completed

> FitHub CRM demonstrates that a Salesforce-based CRM implementation can comprehensively address the operational challenges of a fitness center without writing a single line of traditional application code.

The project was completed across two structured phases: a planning phase establishing scope, objectives, methodology, and risk management, followed by a development phase delivering all four custom objects, lookup relationships, validation rules, a record-triggered automation flow, a custom report, and a live dashboard.

All six modules — Member Management, Trainer Management, Class Management, Booking System, Automation, and Reports & Dashboard — were implemented, tested, and validated. A comprehensive test cycle of six test cases confirmed that every component functions as designed, with all tests returning a PASS result.

The system is production-ready within its defined scope and provides a strong academic demonstration of practical CRM development skills using the Salesforce platform. The identified future enhancements — mobile access, payment integration, advanced analytics, and multi-branch support — outline a clear roadmap for taking this implementation to a commercial-grade deployment.

FitHub CRM — All Phases Complete

System implemented, tested, documented, and submitted for academic evaluation.